

Filing room

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.2 Programming fundamentals — Paper 2, Computational thinking, algorithms and programming
Lesson	Filing room (lesson 3 of the unit)
Lesson ID	pf-l03
Experience ID	pf-l03
Plan ID	PLAN-PF-L03
Content version	a0e570ac1082
Website	https://revise360.co.uk/experience.html?id=pf-l03

Big question, outcomes and success criteria

Big question — How does a program keep hold of data after it has stopped running?

Prerequisites

The lesson before this one is pf-l02. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: pf-l01.

Assessable outcomes

1. **pf-l03-o1** — know why a program needs to store data in a file
2. **pf-l03-o2** — understand how a file is opened, read from, written to and closed
3. **pf-l03-o3** — know the difference between writing to a file and appending to it

Vocabulary, resources and preparation

Key vocabulary — File handling, Append

Resources

SLIDES	presentations/PF_L03_FilingRoom_Lesson.pptx
WORKSHEET	worksheets/PF_L03_FilingRoom_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=pf-l03

Before the lesson

1. Sign in on the **teacher dashboard** (`teacher.html`) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on PF_L03_FilingRoom_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Why files.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Why files

- Data held in variables is lost as soon as the program stops, because variables are held in RAM.
- Writing data to a file stores it on secondary storage, so it is still there the next time the program runs.
- Files also let the output of one program become the input of another.

Challenge on this wall: Give two things a game should save to a file, and say what would happen if it did not.

2. Opening a file

- A file must be opened before a program can read from it or write to it.
- Opening a file gives the program a handle: a name it uses to refer to that file from then on.
- In the OCR Exam Reference Language, `newFile("scores.txt")` creates a new file and `openRead("scores.txt")` opens one that already exists.

Challenge on this wall: Explain what could go wrong if a program tries to open a file that does not exist.

3. Reading a file

- Reading copies data from the file into the program, usually one line at a time.
- A condition-controlled loop reads line by line until the end of the file is reached.
- In the OCR Exam Reference Language, `readLine()` gets the next line and `endOfFile()` becomes True when there are no lines left.

Challenge on this wall: Explain why a program reads a file one line at a time rather than all at once.

4. Writing to a file

- Writing copies data out of the program and into the file, so it is saved on secondary storage.
- In Python 3, `open("scores.txt", "w")` opens a file for writing and deletes anything that was already in it.
- Data written to a file is not safe until the file has been closed.

Challenge on this wall: A program saves the high score using mode "w" every time it runs. Explain the problem this causes.

5. Appending

- Appending adds new data to the end of a file and keeps everything that was already there.
- In Python 3, `open("log.txt", "a")` opens a file in append mode.
- Appending suits a log or a high score table, where the old records must not be lost.

Challenge on this wall: Explain which mode a program should use to add a new score to a file of past scores, and why.

6. Closing the file

- Closing a file tells the operating system that the program has finished with it.
- Data written to a file may still be waiting in a buffer in memory, and closing the file makes sure it reaches the disk.
- A file left open can stop other programs using it, and wastes memory.

Challenge on this wall: Describe two problems that could happen if a program never closed the files it opened.

Retrieval and the check for understanding

Starter — A quiz game keeps a table of high scores. Explain what happens to those scores when the player closes the game, if the program stores them in variables only.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pf - l03 - s4 - c1), so the wording a pupil sees matches the wording you say.

In Python 3, what happens to data already in a file when it is opened with mode "w"?

Correct: It is deleted

What a pupil might choose instead

- It is kept, and the new data is added at the end
- It is kept, and the new data is added at the start
- It is copied into a backup file first

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: Mode "w" starts the file again from empty, so use mode "a" when the old data must be kept.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	13	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	1	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL**: pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND**: the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Why files	2	A file is needed because...
2	Opening a file	2	Opening a file means...
3	Reading a file	2	Reading a file line by line means...
4	Writing to a file	2	Writing to a file means...
5	Appending	2	Appending is different from writing because...
6	Closing the file	2	Closing a file is important because...
final	Final challenge: read the file	2	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pf-l03-x1	State one reason why a program writes data to a file rather than keeping it in a variable.	2	worksheet, Exam practice a); deck, Exam practice slide
pf-l03-x2	Describe the four operations a program carries out on a file, in the order they happen.	4	worksheet, Exam practice b); deck, Exam practice slide
pf-l03-x3	A program records every login in a file called log.txt. Explain why the file should be opened in append mode rather than write mode.	3	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pf-l03.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Why does a program write data to a file?** — So the data can be used inside a loop; So the data takes up less memory
- **A quiz program keeps its high scores in variables only. What happens w** — The high scores are moved to the hard disk automatically; The high scores become constants
- **Why must a file be opened before it is used?** — Files are encrypted until they are opened; Opening a file empties it ready for new data
- **In the OCR Exam Reference Language, what does `newFile("log.txt")` do?** — Opens log.txt ready for reading; Deletes log.txt
- **What does `endOfFile()` give once every line has been read?** — False; The last line of the file

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- Why a file is needed at all
- How to open, read, write and close a file
- The difference between writing to a file and appending to it

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is pf-l04, Record vault. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

Revise 360 — a product of Revise 360 Ltd, created by Olly Pettitt. An independent resource: not written, endorsed or approved by OCR or any exam board. Classroom use and personal study only.